UNIT 1: Function of Complex Variables - Complex Analysis I

- **Introduction:**

→ A number whose square is -1 is called an imaginary or a complex quantity and is denoted by i (iota).

$$i = \sqrt{-1}$$

$$\boxed{i^2 = -1}$$

→ A complex number is written as,

$$\boxed{Z = x + iy} \qquad (x, y \in R).$$

where,

x is real part of Z, $\quad Re(Z) = x$

y is imaginary part of Z, $\quad Im(Z) = y$

1) a) Geometric Representation of Complex Numbers :-

$Z = x + iy$

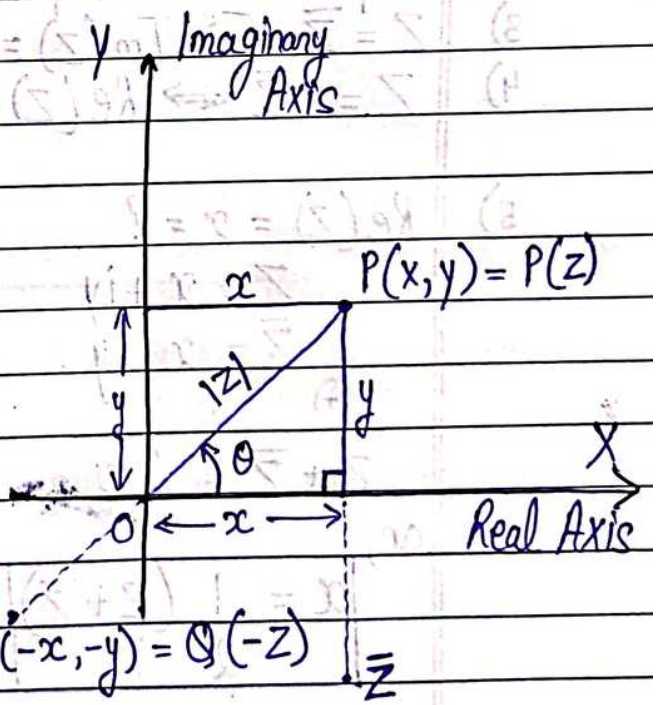
and,

conjugate of $Z = \bar{Z} = x - iy$.

$P(Z) = x + iy$

and,

$Q(-Z) = -x - iy$.



b) Modulus of a Complex Number :-

$$|Z| = \sqrt{x^2 + y^2}.$$

c) Argument of a Complex Number :-

- 'θ' is called argument or amplitude of Z.
- If argument lies in the range of,
  $-\pi < \theta \leq \pi$.
  then, 'θ' is called principal argument.
- General Argument is written as,
  General Argument = Principal Argument $+ 2K\pi$ $(K \in I)$

d) Few Properties :-

1) $\overline{(\overline{z})} = z$
2) $z_1 = z_2 \Leftrightarrow \overline{z_1} = \overline{z_2}$
3) $z = \overline{z} \Leftrightarrow Im(z) = 0$ (Purely Real)
4) $z = -\overline{z} \Leftrightarrow Re(z) = 0$ (Purely Imaginary)

5) $Re(z) = x = ?$

$z = x + iy$

$\overline{z} = x - iy$.

$\oplus \quad \oplus$

$z + \overline{z} = 2x$

or, $\boxed{x = \frac{1}{2}(z + \overline{z})}$.

6) $Im(z) = y = ?$

$z = x + iy$

$\overline{z} = x - iy$

$\ominus \quad \ominus$

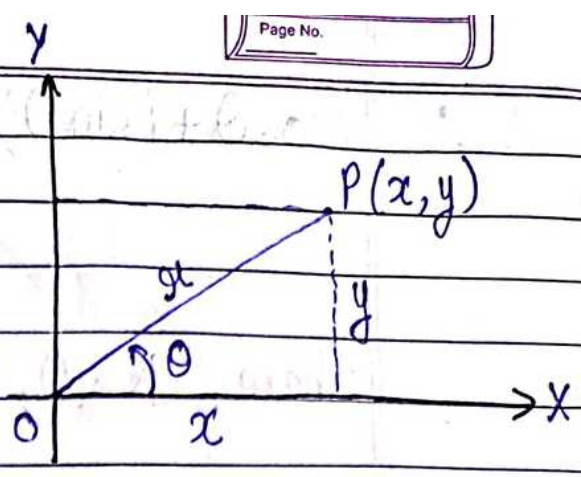
$z - \overline{z} = 2iy$.

or, $\boxed{y = \frac{1}{2i}(z - \overline{z})}$.

e) Polar Form of a Complex Number :-

Cartesian, $P(x, y) \longrightarrow$ Polar form, $Z(r, \theta)$.

$\Rightarrow$ $Z = r[\cos\theta + i\sin\theta]$



where,

- $r = \sqrt{x^2+y^2}$.

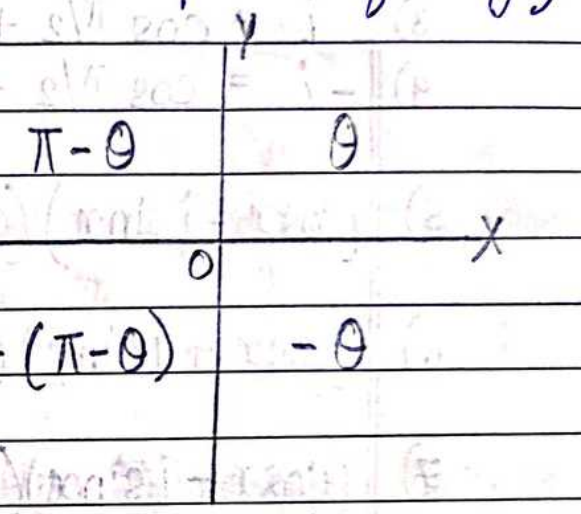
and,

- $\theta = \tan^{-1}\left|\frac{y}{x}\right|$.

(NOTE: 'θ' will depend on coordinate axes / values of x & y)

Note: (TRICK to determine 'θ'):-

Quadrant		Principal Argument
1) I	$\rightarrow$	θ
2) II	$\rightarrow$	$\pi-\theta$
3) III	$\rightarrow$	$-(\pi-\theta)$
4) IV	$\rightarrow$	$-\theta$



2) De-Moivre's Theorem :-

Let, $Z = r(\cos\theta + i\sin\theta)$

Then,

$$\boxed{Z^n = r^n(\cos n\theta + i\sin n\theta)}$$

where,

n is any positive or negative integer or a real rational number,

$n \in I$ and $n \in Q$.

2.1) Roots of a Complex Number :-

In general,

$\cos\theta = \cos(2k\pi+\theta)$, where $k \in I$

$\sin\theta = \sin(2k\pi+\theta)$, where $k \in I$.

$$\therefore (\cos\theta + i\sin\theta)^{1/n} = [\cos(2k\pi+\theta) + i\sin(2k\pi+\theta)]^{1/n}$$

$$= \cos\left(\frac{2k\pi+\theta}{n}\right) + i\sin\left(\frac{2k\pi+\theta}{n}\right)$$

where, $k = 0, 1, 2, 3, \ldots, (n-1)$.

Note:

1) $1 = \cos 0 + i\sin 0$
2) $-1 = \cos\pi + i\sin\pi$
3) $i = \cos\pi/2 + i\sin\pi/2$
4) $-i = \cos\pi/2 - i\sin\pi/2$.

} (Standard numbers).

5) $(\cos x + i\sin x)(\cos y + i\sin y) = \cos(x+y) + i\sin(x+y)$.

6) $(\cos x + i\sin x)(\cos y - i\sin y) = \cos(x-y) + i\sin(x-y)$.

7) $(\cos x - i\sin x)(\cos y + i\sin y) = \cos(x-y) - i\sin(x-y)$

8) $(\cos x - i\sin x)(\cos y - i\sin y) = \cos(x+y) - i\sin(x+y)$

3)(i) **Function of Complex Variables :-**

$\Rightarrow f(z) = u + iv = u(x,y) + i\,v(x,y)$.

(Real part) (Imaginary part)

or, $\boxed{w = u + iv}$

Here,
$f(z)$ is a function of complex variable and is denoted by

(ii) **Continuity of function $f(z)$:-**

$\Rightarrow$ A function $w = f(z)$ is said to be continous at $z = z_0$ if,

$$\boxed{\lim_{z \to z_0} f(z) = f(z_0)}$$

(iii) Differentiability of function $f(z)$:-

⇒ Let, $f(z)$ be a singled value function of the variable z, then

$$f'(z) = \lim_{\Delta z \to 0} \frac{f(z+\Delta z) - f(z)}{\Delta z}$$

- Provided that the limit exists and is independent of path along $\Delta z \to 0$.

Singled Valve Function : Any function that gives only one output valve for one input valve is a singled valve function Examples are, $f(x) = x$. For $x = n$, $n \in R$, there will be always one valve of $f(x)$... —1.

4) Analytic Function : (Regular Functions OR Hylomorphic Functions)

⇒ A singled valued function $f(z)$ which is differentiable at $z = z_0$ is said to be analytic function at the point $z = z_0$ (Point z_0 is said non-singular point or regular).

- Singular Point → The point at which the function is not differentiable is called a singular point of the function.

⇒ The Necessary Condition of $f(z)$ to be analytic : (at all points in a region R).

① $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$ → Provided $\partial u/\partial x$, $\partial v/\partial y$, $\partial u/\partial y$, $\partial v/\partial x$ exist

② $\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$ → C-R Equations. (Cauchy - Reimann Equations)

★ Sufficient Condition for $f(z)$ to be Analytic,

$$f'(z) = \frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x} \longrightarrow (1)$$

OR

$$f'(z) = \frac{\partial v}{\partial y} - i\frac{\partial u}{\partial y} \longrightarrow (2)$$

★ • C-R Equations in Polar Form :-

$$\frac{\partial u}{\partial r} = \frac{1}{r}\frac{\partial v}{\partial \theta} \quad \text{and,} \quad \frac{\partial u}{\partial \theta} = -r\frac{\partial v}{\partial r}$$

• Derivative of $w = f(z)$ in Polar form :-

i) $$\frac{\partial w}{\partial z} = (\cos\theta - i\sin\theta)\frac{\partial w}{\partial r} \quad \text{—①}$$

ii) $$\frac{\partial w}{\partial z} = \frac{-i}{r}(\cos\theta - i\sin\theta)\frac{\partial w}{\partial \theta} \quad \text{—②}$$

4.1) Harmonic Functions :-

• Any function of x and y (of first and second order) if satisfies Laplace equation is called Harmonic function.

★ For example,

(For 'u') $$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

(For 'v') $$\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = 0$$

} Laplace Equation.

★ Laplace Equations and Conjugates :-

$$\boxed{\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0} \qquad \boxed{\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = 0}$$

- **How to find 'v' if 'u' is given →**

If, $u(x,y)$ is given. Then,

$$\boxed{du = \frac{\partial u}{\partial x}\,dx + \frac{\partial u}{\partial y}\,dy}$$

and, $f(z)$ is analytic i.e.,

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}.$$

So,

$$\boxed{du = \frac{\partial v}{\partial y}\,dx - \frac{\partial v}{\partial x}\,dy}$$ (exact differential form) (For 'u(x,y)')

Similarly,

$$\boxed{dv = -\frac{\partial u}{\partial y}\,dx + \frac{\partial u}{\partial x}\,dy}.$$ (For 'v(x,y)')

4.2) **Milne's Thomson Method →**

If $u(x,y)$ is given,

$$f(z) = \int [\phi_1(z,0) - i\,\phi_2(z,0)]\,.\,dz$$

Here,

$$\frac{\partial u}{\partial x} = \phi_1(x,y), \quad \frac{\partial u}{\partial y} = \phi_2(x,y)$$

If $v(x,y)$ is given,

$$f(z) = \int [\psi_1(z,0) + i\,\psi_2(z,0)]\,dz$$

Here,

$$\frac{\partial v}{\partial y} = \psi_1(x,y) \quad , \quad \frac{\partial v}{\partial x} = \psi_2(x,y)$$

5)(i) Exponential Function of Complex Variable →

⇒ Exponential function of complex variable $Z = x+iy$ is written as, e^z or $\exp Z$ and, defined as

$$\Rightarrow e^z = 1 + z + \frac{z^2}{2!} + \frac{z^3}{3!} + \ldots\ldots + \frac{z^n}{n!} + \ldots \infty$$

⇒ PROPERTIES →

(1) ⇒ Exponential form of $Z = re^{i\theta}$

⇒ $Z = x+iy = r(\cos\theta + i\sin\theta) = re^{i\theta}$.

(2) → e^z is periodic function having imaginary period $2\pi i$

(3) → e^z is not zero for any value of z.

(ii) Euler's formula →

If θ be real or complex,

$$\boxed{e^{i\theta} = \cos\theta + i\sin\theta} \quad \text{OR} \quad \boxed{e^{-i\theta} = \cos\theta - i\sin\theta}$$

→ Adding ① and ②,

$$\Rightarrow e^{i\theta} + e^{-i\theta} = 2\cos\theta$$

or,

$$\boxed{\cos\theta = \frac{e^{i\theta} + e^{-i\theta}}{2}}$$

→ Similarly, subtracting eq-② from eq-①,

$$\sin\theta = \frac{1}{2i}\left(e^{i\theta} + e^{-i\theta}\right)$$

or,

$$\boxed{\sin\theta = \frac{1}{2i}\left(e^{i\theta} + e^{-i\theta}\right)}$$

And,

$$\boxed{\tan\theta = \frac{e^{i\theta} - e^{-i\theta}}{i\left(e^{i\theta} + e^{-i\theta}\right)}}$$

(iii) Hyperbolic Functions → For values of x (real or complex), the quantity, $\frac{e^x + e^{-x}}{2}$ is called hyperbolic cosine of x and is written as $\cosh x$.

And,

$$\sinh x = \frac{e^x - e^{-x}}{2}$$

★ Relation between hyperbolic and circular functions :-

We know that,

$$\sin\theta = \frac{e^{i\theta} - e^{-i\theta}}{2}, \quad \cos\theta = \frac{e^{i\theta} + e^{-i\theta}}{2}$$

Putting $\theta = ix$,

$$\Rightarrow \sin ix = \frac{e^{i(ix)} - e^{-i(ix)}}{2i} = \frac{e^{-x} - e^{x}}{2i} \times \frac{i}{i}$$

$$= \frac{-i^2(e^x - e^{-x})}{2i} = i\left(\frac{e^x - e^{-x}}{2}\right)$$

$$= i\sinh x.$$

$$\boxed{\sin ix = i \sinh x}$$

Similarly,

$$\boxed{\cos ix = \cosh x}$$

$$\boxed{\tan ix = i \tanh x}$$

- To seperate the real and imaginary part of $\sin(x+iy)$.

$$\Rightarrow \sin(x+iy) = \sin x \cdot \cos iy + \cos x \cdot \sin iy$$
$$= \underbrace{\sin x \cdot \cosh y}_{\text{Real part}} + \underbrace{i \cos x \cdot \sinh y}_{\text{Imaginary part}}.$$

- To seperate the real and imaginary part of $\tan(x+iy)$

$$\Rightarrow \tan(x+iy) = \frac{\sin(x+iy)}{\cos(x+iy)} \times \frac{\cos(x-iy)}{\cos(x-iy)} \times \frac{2}{2}$$

$$= \frac{2\sin(x+iy)\cdot\cos(x-iy)}{2\cos(x+iy)\cdot\cos(x-iy)}$$

$$= \frac{\sin(x+iy+x-iy) + \sin(x+iy-x+iy)}{\cos(x+iy+x-iy) + \cos(x+iy-x+iy)}$$

$$= \frac{\sin 2x + \sin(2iy)}{\cos 2x + \cos(2iy)}$$

$$\Rightarrow \tan(x+iy) = \frac{\sin 2x + i \sinh 2y}{\cos 2x + \cosh 2y}$$

On seperating,

→ Real part = $\frac{\sin 2x}{\cos 2x + \cosh(2y)}$

→ Imaginary part = $i\,\frac{\sinh 2y}{\cos 2x + \cosh 2y}$.

(iv) Logarithm of Complex Numbers :

⇒ $Z = x + iy = r(\cos\theta + i\sin\theta) = re^{i\theta}$

⇒ $x + iy = re^{i\theta}$

or,

⇒ $\log(x+iy) = \log(re^{i\theta}) = \log r + i\theta$.

$\because r = \sqrt{x^2+y^2}$ and, $\tan^{-1}\left(\frac{y}{x}\right) = \theta$

⇒ $\boxed{\log(x+iy) = \frac{1}{2}\log(x^2+y^2) + i\tan^{-1}\left(\frac{y}{x}\right)}$

→ $\underline{\log}(x+iy)$ → Principal value.

→ $\underline{\text{Log}}(x+iy)$ → General value. $(n \in N)$

$\therefore \text{Log}(x+iy) = \frac{1}{2}\log(x^2+y^2) + i\left[\tan^{-1}\left(\frac{y}{x}\right) + 2n\pi\right]$.

6) Line Integral in the Complex Plane :-

⇒ $\int_C f(z)\,dz = \int_C (u+iv)(dx+i\,dy)$ (C → closed integral)

Here,

⇒ $Z = x + iy$

⇒ $f(z) = u + iv = u(x,y) + iv(x,y)$

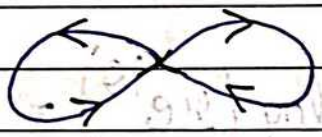
and,

⇒ $dZ = dx + idy$

7) Cauchy's Integral Theorem and Cauchy's Integral Formula :-

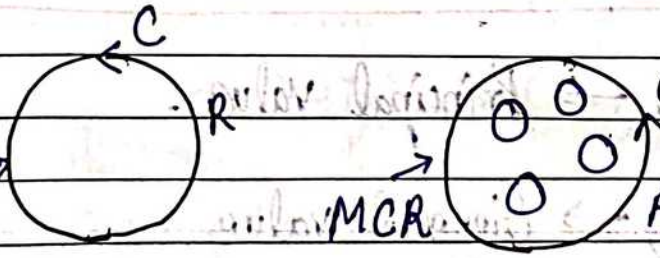
⇒ A simple closed curve is a curve that does not cross itself.

(a) Multiple curve : A curve which crosses itself is called a multiple cure.



(b) Simply connected region : A simply connected region is one which has no pole (holes).

(c) Multiple or Multiply connected region : A multiple connected region is one which has more than one poles (holes).



Note : We can convert multiply connected region into simply connected region with the help of cross-cuts.

7.1) Cauchy's Integral Theorem statement :

⇒ If a function $f(z)$ is analytic and its derivative $f'(z)$ continous at all points inside and on a simple closed curve C, then;

$$\boxed{\int_C f(z).\,dz = 0}$$

$$\Rightarrow \int_C F(z).\,dz = \int_C (u+iv)(dx+idy) \qquad \left[\begin{array}{l} z = x+iy. \\ \text{So,} \\ f(z) = u+iv \end{array}\right].$$

$$= \int_C u.\,dx + iu\,dy + iv\,dx - v.\,dy$$

$$= \int_C (u\,dx - v\,dy) + i\,(u\,dy + v\,dx)$$

$$= \int_C \left[(u\,dx - v\,dy) + i\,(v\,dx + u\,dy)\right].$$

$$= \int_C (u\,dx - v\,dy) + i\int_C (v\,dx + u\,dy)$$

Note: (Green's Theorem) →

$$\int_C \phi\,dx + \psi\,dy = \iint_R \left(\frac{\partial}{\partial x}\psi - \frac{\partial}{\partial y}\phi\right) dx.dy.$$

$$= \iint \left(-\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}\right) dx.dy + i\iint \left(\frac{\partial u}{\partial x} - \frac{\partial v}{\partial y}\right) dx\,dy.$$

$\because$ $f(z)$ is analytic, i.e.,

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}.$$

$$= \iint \left(\frac{\partial v}{\partial y} - \frac{\partial v}{\partial y}\right) dx\,dy + i\iint \left(\frac{\partial v}{\partial y} - \frac{\partial v}{\partial y}\right) dx\,dy$$

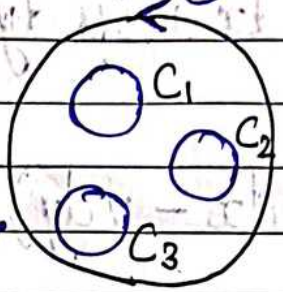
$\Rightarrow \int_C F(z).dz = 0 + i(0) = \underline{0}$

(Hence proved).

7.2) Extension of Cauchy's theorem to multiple connected region

$\Rightarrow \int_C f(z).dz = \int_{C_1} f(z).dz + \int_{C_2} f(z).dz + \int_{C_3} f(z).dz \ldots$

$\int_C f(z).dz = 0$



7.3) Cauchy's Integral formula :-

Statement: If $f(z)$ is analytic function within and on a closed curve C and if 'a' is any point within C, then

$$\boxed{f(a) = \frac{1}{2\pi i} \int_C \frac{f(z)}{(z-a)} dz}$$

7.4) Cauchy's Integral formula for the derivative of an analytic function :-

$\Rightarrow$ If a function $f(z)$ is analytic in a region R, then its derivative at any point $z=a$, if R is also analytic in R, and is, given as;

$$\boxed{f^n(a) = \frac{n!}{2\pi i} \int_C \frac{f(z)}{(z-a)^{n+1}} dz}$$

If $n = 1$;

$$f'(a) = \frac{1!}{2\pi i}\int_C \frac{f(z)}{(z-a)^2}\,dz.$$

If $n=2$,

$$f''(a) = \frac{2!}{2\pi i}\int_C \frac{f(z)}{(z-a)^3}\,dz.$$

8) Expansion of functions :-

① Taylor series
② Maclaurin series
③ Laurents series

8.1) Taylor Series : If a function $f(z)$ is analytic inside a circle C with centre at $z=a$ then it can be expanded in the series,

$$f(z) = f(a) + (z-a)f'(a) + \frac{(z-a)^2}{2!}f''(a) + \dots + \frac{(z-a)^n}{n!} \times f^n(a) + \dots$$

⇒ If putting $a=0$,

⇒ $$f(z) = f(0) + Z f'(0) + \frac{Z^2}{2!}f''(0) + \dots + \frac{Z^n}{n!}f^n(0) + \dots$$

This series is called Maclaurin series.

8.2) Laurent's Series :-

(co-centric)

If $f(z)$ is analytic in annular region R bounded by two concentric circles C_1 and C_2 of radii r_1 and r_2 ($r_1 \geq r_2$) and centre at 'a', then for all z in R, we have;

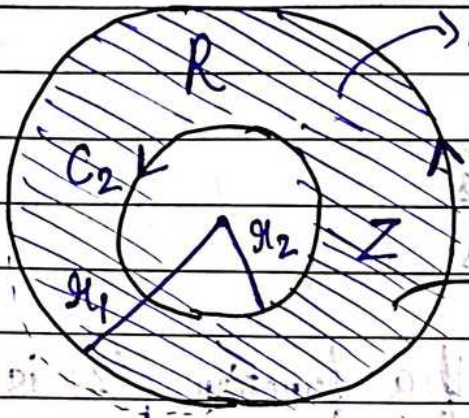
$$f(z) = \sum_{n=0}^{\infty} a_n (z-a)^n + \sum_{n=1}^{\infty} b_n (z-a)^{-n}$$

(General term) (Principal term)

where,

$$a_n = \frac{1}{2\pi i}\int_c \frac{f(t)}{(t-a)^{n+1}}\,dt \quad , n = 0, 1, 2, 3, \ldots$$

$$b_n = \frac{1}{2\pi i}\int_c \frac{f(t)}{(t-a)^{-n+1}}\,dt \quad , n = 1, 2, 3, \ldots$$



9) Power Series: A series in powers of $(z-z_0)$ is called pa
Series:

$$\sum_{n=0}^{\infty} a(z-z_0)^n = a_0 + a_1(z-z_0) + a_2(z-z_0)^2 + \ldots$$

$$\ldots + a_n(z-z_0)^n$$

Note: If centre of series, i.e., $z_0 = 0$, then;

$$\sum_{n=0}^{\infty} a_n z^n = a_0 + a_1 z + a_2 z^2 + a_3 z^3 + \ldots$$

9.1) Radius of Convergence of Power Series :-

⇒ Let the power series, $\sum a_n z^n$, by cauchy theorem on limits

⇒ Radius of convergence (R) is given by,

$$\frac{1}{R} = \lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| .$$

10) Zero of Analytic Function :-

→ A zero of analytic function $f(z)$ is the value of z for which $f(z) = 0$, i.e.,

$f(z) = z - 2$

So for $z = 2$, $f(z) = 0$.

11) • Singular Point :-

⇒ A point at which a function $f(z)$ is not analytic is known a singular point or singularity of the function.

Let, $f(z) = \frac{1}{z-2}$ has a singular point at $z = 2$.

• Isolated Singular Point :-

⇒ If $z = a$ is a singularity of $f(z)$ and if there is no other singularity with in a small circle surrounding the point $z = a$, then $z = a$ is said to be an isolated singularity of function $f(z)$, otherwise it is called non-isolated singular point.

Example: ① $f(z) = \frac{1}{(z-1)(z-3)}$

Here, $z = 1, 3$ are two isolated singular point.

② $f(z) = \frac{1}{\sin(\pi/z)}$

here, $\sin \pi/z = 0$

$\Rightarrow \pi/z = n\pi$.

$\Rightarrow z = 1/n$.

Here, $z = \frac{1}{n}$ $(n = 0, 1, 2, 3, \ldots \infty)$ $\therefore z = 1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \ldots$ $(n \in N)$

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- **Pole of order 'm' :-**

→ By Laurent's series,

$$\Rightarrow f(z) = \sum_{n=0}^{\infty} a_n (z-z_0)^n + \sum_{n=1}^{\infty} b_n (z-z_0)^{-n}$$

$$\hookrightarrow f(z) = a_0 + a_1(z-z_0) + \dots + \frac{b_1}{(z-z_0)} + \frac{b_2}{(z-z_0)^2} + \dots + \frac{b_m}{(z-z_0)^m} + \dots + \frac{b_{m+1}}{(z-z_0)^{m+1}} + \dots \quad \textcircled{1}$$

$\Rightarrow$ If the number of terms of negative powers in ① is infinite then, $z = z_0$ is called an essential singularity point of $f(z)$.

- **Removable Singularity :-**

Note → If Laurent's series does not contain negative power of $(z-z_0)$ then, $z = a$ is called a removable singularity.

Example: $f(z) = \dfrac{\sin(z-a)}{(z-a)}$

$$\Rightarrow f(z) = \frac{1}{(z-a)}\left[(z-a) - \frac{(z-a)^3}{3!} + \frac{(z-a)^5}{5!} - \dots \infty\right]$$

(Expansion of $\sin x$)

$$\Rightarrow f(z) = 1 - \frac{(z-a)^2}{3!} + \frac{(z-a)^4}{5!} - \dots \infty$$

→ Now, there is no singularity.

Note: ① If $\lim_{z \to a} f(z)$ = Exists and finite, then $z = a$ is a removable singular point.

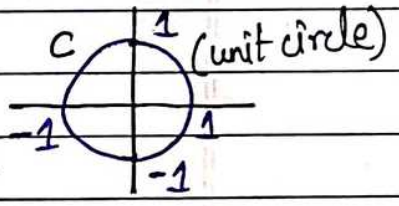
② If $\lim_{z \to a} f(z)$ does not exists, then $z = a$ is an essential singular point.

③ If $\lim_{z \to a} f(z)$ is infinite then $f(z)$ has pole at $z = a$

12) Evaluation of real definite integrals by Contour Integration

1) Integration round the unit circle of the type,

$$\int_0^{2\pi} f(\cos\theta, \sin\theta) \cdot d\theta .$$



here, $\int_C \rightarrow$ Unit circle.

To change $\theta \rightarrow z$, we have to substitute $z = e^{i\theta}$

$\Rightarrow z = e^{i\theta}$

$\Rightarrow dz = ie^{i\theta} \cdot d\theta .$

$\Rightarrow dz = iz \cdot d\theta \quad \Rightarrow d\theta = \frac{dz}{iz} .$

$\Rightarrow \cos\theta = \frac{e^{i\theta} + e^{-i\theta}}{2}$

$\Rightarrow \cos\theta = \frac{z + z^{-1}}{2} = \frac{z^2 + 1}{2z}$

and,

$\Rightarrow \sin\theta = \frac{e^{i\theta} - e^{-i\theta}}{2i} = \frac{z - z^{-1}}{2i} = \frac{z^2 - 1}{2iz} .$